



Rising Researchers Presents

The Undergraduate Research Survival Guide

For Computer Science



Today's Agenda

I

CS
x
Research

II

Finding
My Lab

III

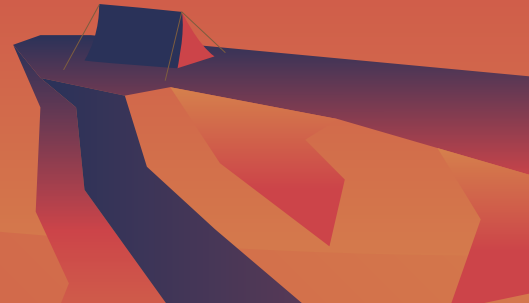
What Happens
Now?

IV

Myth or Fact?

Rising Researchers X Bit Camp

- Student organization
- Aim to teach UMD students about academic research through workshops and modules
- Equip members and audiences with tools for success in a variety of STEM careers





1

CS x Research

How do Computer Science and Research Come Together?

CS x Research

Theoretical
Research

Applied
Research

Tooling
and Data
Science

Conferences > Proceedings 35th Annual Sympo... ?

A new efficient radix sort

Publisher: IEEE

Cite This

PDF



Blockchain: Research and Applications

Available online 29 May 2022, 100092
In Press, Journal Pre-proof



Research Article

Blockchain-based covid vaccination registration
and monitoring

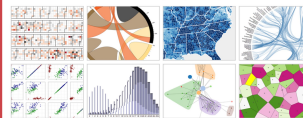
Article | [Open Access](#) | [Published: 18 June 2019](#)

Functional characterization of retinal ganglion cells
using tailored nonlinear modeling

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshi@gmx.com
www.bitcoin.org

D³: Data-Driven Documents
Michael Bostock, Vadim Osherson and Jeffrey Heer



Common Computer Science Research Areas

**Security &
Privacy**

**Language
Processing**

**Artificial
Intelligence**

**Machine
Learning**

Robotics

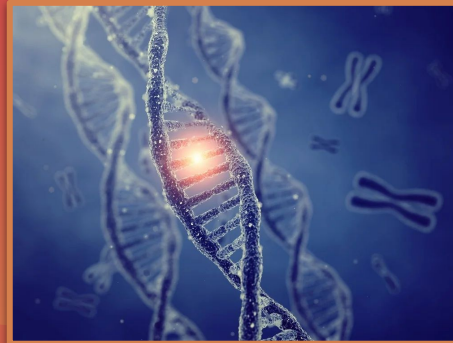
**Vision &
Graphics**

**Databases &
Data Mining**

**Quantum
Computing**

Discuss!

How Do You Think CS Can be Applied in These Fields?



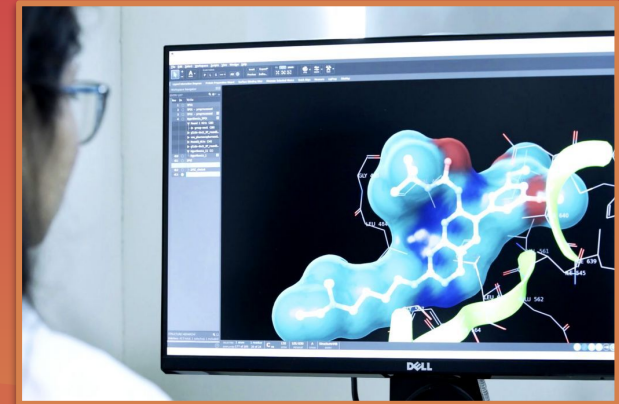
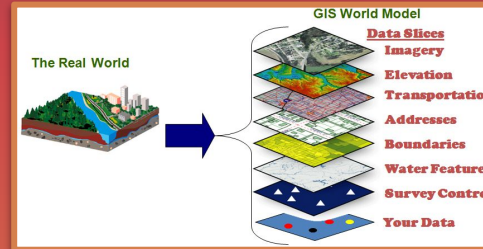
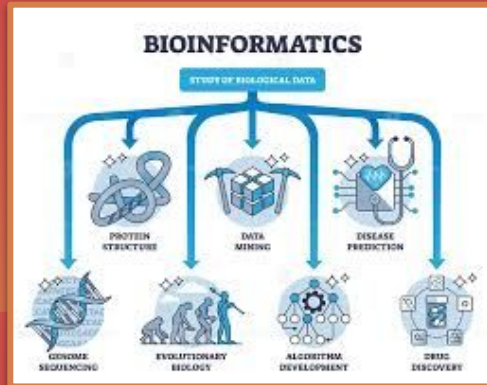
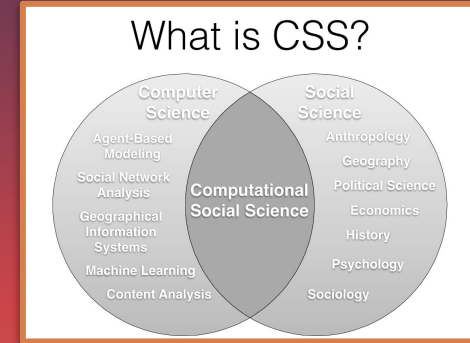
CS is Very Applicable



CellPress **Neuron**

NeuroView
The next generation of neuroscientists needs to learn how to code, and we need new ways to teach them

Ashley L. Juavinett^{1,*}
¹Division of Biological Sciences, Neurobiology Section, 9500 Gilman Drive, La Jolla, CA 92093, USA
*Correspondence: ajuvinn@ucsd.edu
<https://doi.org/10.1016/j.neuron.2021.12.001>





2

Finding My Lab

Finding Your First Research Opportunity!

Finding Campus Opportunities

Professors

Almost all professors at UMD do research, reach out to your professors about their labs!

UMD Research Database

Online Database that connects undergrads with professors and opportunities!

Cold Emailing

Department websites have lab descriptions and emails of the Primary Investigators for each of those labs.

How Do I Decide What to Research?

Your Classes

Think about what topics in your classes have been most interesting to you!

Read Papers

Find a few labs that sound cool and read a paper or two to see if their work is interesting to you.

Think of Your Strengths and Passions

Think of what you excel at in your classes.

Cold Emailing: You Try!

Pretend you're emailing the PI of a lab you want to join:
What would you include?



Pretend you're emailing the PI of a lab you want to join:
What would you include?

Writing A Cold Email

Start with some background research

- Research the professor and their lab
- Read the abstracts of at least 1 or 2 recent papers
- Read about any current projects of theirs



Dr. Pup
**Department of Computer
Science University of Maryland**
**Machine Learning, Computer
Vision**

TITLE	CITED BY	YEAR
Imagenet classification with deep convolutional neural networks A Krizhevsky, I Sutskever, GE Hinton Communications of the ACM 60 (6), 84-90	117160	2017
Dropout: a simple way to prevent neural networks from overfitting N Srivastava, G Hinton, A Krizhevsky, I Sutskever, R Salakhutdinov The journal of machine learning research 15 (1), 1929-1958	38957	2014

Current Projects

**Using deep convolutional neural
networks to improve enemy/friend
(cat/dog) classifiers.**

Writing A Cold Email

Start by introducing yourself!

- Name
- University
- Year/Class

To: drpup@umd.edu

Subject: Undergraduate Research Opportunity

Hello Dr. Pup,

**My name is Maia and I am a first-year
Computer Science Major at UMD!**

Writing A Cold Email

Briefly mention your interests and what led you to reaching out

- Fields you are interested in
- How you learned about the professor or the opportunity
- Why you are reaching out

You can even ask to set up a time to meet before you delve into specifics

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Hello Dr. Pup,

My name is Maia and I am a first-year Computer Science Major at UMD!

I recently came across your profile on Google Scholar and noticed that you are currently pursuing research in Machine Learning and Computer Vision. I would love to get more experience with these fields and wanted to know if you would be available to meet to discuss your current research.

Writing A Cold Email

Dive into specifics to showcase your interest!

- Papers you read
- Current projects you are interested in

■ Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0%, respectively, which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully connected layers we employed a recently developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

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I read your paper on Image Net Classification using Deep Convolutional Neural Networks and was amazed by the results your model had achieved on the ImageNet contest. I'm particularly interested in how you are planning on applying this model to the "enemy/friend classification" problem you had mentioned on your website.

Writing A Cold Email

**If you have experience,
mention it!**

- Relevant courses
- Previous personal/research projects
- Previous work experiences

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Hello Dr. Pup,

My name is Maia and I am a first-year Computer Science Major at UMD! I recently came across your profile on Google Scholar and noticed that you are currently pursuing research in Machine Learning and Computer Vision. **Last semester, I took a course on Machine Learning**, and I would love to get more experience with these fields and wanted to know if you would be available to meet to discuss your current research.

I read your paper on Image Net Classification using Deep Convolutional Neural Networks and was amazed by the results your model had achieved on the ImageNet contest. **I've been training Deep Convolutional Neural Networks in class for multiple toy applications, but would like to work on solving a real-world problem with DCNNs.** I'm particularly interested in how you are planning on applying this model to the "enemy/friend classification" problem you had mentioned on your website.

Writing A Cold Email

Reiterate that you want to meet to discuss their research and potential research work

- Add a question so that the professor has a direct action-item to respond to
- Provide a potential time frame
- Mention that you are *looking to get involved*

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Would you have any availability to meet next week? I am free between 12 and 2pm M-F. I'd love to get a chance to discuss your research and also get your advice on how I can get involved with ML/CV research!

Writing A Cold Email

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Thank you for your time,
Emily

3

Working in the Lab

What do I do after I've joined the lab?
How do I make the most of my time?

What Happens Now?

You've found your brand-new research home, and now it's time to get into the work. But what exactly does that mean?

Learning >> Producing

Explore projects in the lab (even off-duty)

Ask lots of questions (to everyone in the lab)

Reflect on your own curiosities ~ what do you want to do?



Things to do Besides Research

- Attend talks
- Connect with other undergraduate research peers!
- Reflect often about what you learn and what sparks your curiosity
- Get to know your lab ~ spending time in the lab, even without having specific duties, can help you learn more too!

Take Care of Yourself, Too!

*The process of research can be one that is non-linear and, at times, redundant, as a project takes on a life of its own. **Patience is key**, but burnout is understandable.*

- Take breaks
- Ask for (non-science related) advice!
- Plan out your research milestones, but allow for change
- Reflect and see how far you've come
- Ask your mentor for advice ~ a good mentor will be supportive!
- Keep an eye on the big picture



4

Myth of Fact?

Let's Play a Game!

Myth or Fact: Mentimeter





Ask us Anything!

AND KEEP IN TOUCH!



@risingresearchersumd on Instagram!



risingresearchers01@gmail.com



go.umd.edu/risingresearchers

**Find Some Research
Opportunities Here!**

